Gradient Descent

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Gradient descent

Definition 0.1 (Gradient descent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive **step lengths**. **Gradient descent** [1] computes, for $t = 0, 1, 2, \ldots$,

$$\underset{\sim}{x}^{(t+1)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t)} - s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

Example 0.1 (Fixed steps on a parabola)**.** For $f(x) = x^2$, $f'(x) = 2x$, so gradient descent with fixed step length s is

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - 2s\, x^{(t)} \\ &= (1 - 2s)\, x^{(t)}, \end{aligned}$$

and $x^{(t)} = (1 - 2s)^t x^{(0)}$.

- With $s = 0.25$, $x^{(t)} = 0.5^t x^{(0)}$, which tends to the minimizer 0.
- With $s = 0.5$, the first step lands on 0, since $1 - 2s = 0$.
- With $s = 1.5$, $x^{(t)} = (-2)^t x^{(0)}$, which moves away from 0 for any $x^{(0)} \neq 0$: too long a step can make gradient descent fail.

Remark (Gradient descent as Newton's method with a simpler Hessian)

Remark 0.1 (Gradient descent as Newton's method with a simpler Hessian). Gradient descent (Definition 0.1) is Newton's method (Definition 5 in Newton's Method) with the Hessian replaced by $\frac{1}{s^{(t)}} \mathbf{I}_p$, so it needs no second derivatives.

Definition 0.2 (Gradient ascent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive step lengths. **Gradient ascent** [1] repeats, for $t = 0, 1, 2, \ldots$, the update

$$\underset{\sim}{x}^{(t+1)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t)} + s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

Example 0.2 (Gradient ascent is gradient descent on the negative function)**.** Gradient ascent on f takes the same steps as gradient descent on $-f$, since $\frac{\partial}{\partial x}(-f) = -\frac{\partial}{\partial x} f$. In one variable, for $f(x) = -x^2$ the derivative is $f'(x) = -2x$, so gradient ascent with fixed step length s is $x^{(t+1)} = (1 - 2s)\, x^{(t)}$, the iteration of Example 0.1, which tends to the maximizer 0 when $0 < s < 1$.

Definition 0.3 (Line search)**.** In gradient descent (Definition 0.1), a **line search** [2] is a rule that chooses the step length $s^{(t)}$ by examining the values of f along the line $\left\{\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}) : s > 0\right\}$.

Definition 0.4 (Exact line search)**. Exact line search** is the line search (Definition 0.3) that chooses $s^{(t)}$ to minimize $f(\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}))$ over $s > 0$.

Example 0.3 (Exact line search on a parabola)**.** For $f(x) = x^2$ and $x^{(t)} = 2$, the derivative is $f'(2) = 4$, so the line is $\{2 - 4s : s > 0\}$ and the values along it are

$$\begin{aligned} \phi(s) &= f(2 - 4s) \quad \text{(the value at step length } s\text{)} \\ &= (2 - 4s)^2 \quad (f(x) = x^2) \end{aligned}$$

$\phi(s) \geq 0$ for every s, and

$$\begin{aligned} \phi(0.5) &= (2 - 2)^2 \\ &= 0, \end{aligned}$$

so exact line search chooses $s^{(t)} = 0.5$, which lands on the minimizer $x^{(t+1)} = 0$ in one step.

Definition 0.5 (Descent direction)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have a gradient $\underset{\sim}{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$ at $\underset{\sim}{x}$. A direction $\underset{\sim}{d}$ (direction) is a **descent direction** for f at $\underset{\sim}{x}$ if $\underset{\sim}{g} \cdot \underset{\sim}{d} < 0$. When $\underset{\sim}{g} \neq \underset{\sim}{0}_{p \times 1}$, $-\underset{\sim}{g}$ is a descent direction, since $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -\|\underset{\sim}{g}\|^2 < 0$.

Example 0.4 (Descent directions for $x_1^2 + x_2^2$ at $(1,0)^\top$)**.** Let $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ and $\underset{\sim}{x} = (1,0)^\top$, so

$$\begin{aligned}\underset{\sim}{g} &= (2x_1,\ 2x_2)^\top \\ &= (2,0)^\top.\end{aligned}$$

- $\underset{\sim}{d} = (-1,1)^\top$ is a descent direction:

$$\begin{aligned}\underset{\sim}{g} \cdot \underset{\sim}{d} &= -2 + 0 \\ &= -2 \\ &< 0.\end{aligned}$$

- $\underset{\sim}{d} = (0,1)^\top$ is not: $\underset{\sim}{g} \cdot \underset{\sim}{d} = 0$.
- $-\underset{\sim}{g} = (-2,0)^\top$ is: $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -4 < 0$.

Theorem 0.1 (A short enough step downhill decreases the function)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$ (continuity), and let $\underset{\sim}{x} \in \mathbb{R}^p$ be a point with $\underset{\sim}{g} \overset{def}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \neq \underset{\sim}{0}_{p\times 1}$. Then there is an $\bar{s} > 0$ with $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$ for every $s \in (0, \bar{s})$: a short enough step in the descent direction $-\underset{\sim}{g}$ (Definition 0.5) decreases f.*

Proof

Proof. Let $h(s) \overset{\text{def}}{=} f(\underset{\sim}{x} - s\underset{\sim}{g})$. The inner function $s \mapsto \underset{\sim}{x} - s\underset{\sim}{g}$ has derivative $-\underset{\sim}{g}^\top$ (vector-valued derivative), so by the vector chain rule (vector chain rule)

$$\begin{aligned}
h'(s) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x} - s\underset{\sim}{g}), && \text{(chain rule)} \\
h'(0) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) && \text{(set } s = 0\text{)} \\
&= -\underset{\sim}{g} \cdot \underset{\sim}{g} && \text{(definition of } \underset{\sim}{g}\text{)} \\
&= -\|\underset{\sim}{g}\|^2 && \text{(squared length)} \\
&< 0. && (\underset{\sim}{g} \neq \underset{\sim}{0}_{p\times 1})
\end{aligned}$$

Since $h'(0) = \lim_{s\to 0}(h(s) - h(0))/s$ is negative, taking $\varepsilon = |h'(0)|$ in the definition of the limit gives an $\bar{s} > 0$ such that $(h(s) - h(0))/s < h'(0) + |h'(0)| = 0$ for $0 < s < \bar{s}$. Multiplying by $s > 0$, $h(s) < h(0)$, that is, $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$. □

Example 0.5 (How short is short enough)**.** In Example 0.1, $f(x) = x^2$ at $x \neq 0$ has $g = 2x$, and $f(x - 2sx) = (1-2s)^2x^2 < x^2$ exactly when $|1-2s| < 1$, that is, when $0 < s < 1$. So Theorem 0.1 holds with $\bar{s} = 1$, and the step $s = 1.5$ of that example was too long.

Example 0.6 (Gradient descent zigzags)**.** Let $f(\underset{\sim}{x}) = \frac{1}{2}(x_1^2 + b\,x_2^2)$ with $b > 0$, so $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = (x_1,\ bx_2)^\top$ and one gradient step of length s from $\underset{\sim}{x}$ gives $((1-s)x_1,\ (1-bs)x_2)^\top$. For exact line search (Definition 0.4), let $\phi(s) \overset{\text{def}}{=} f$ at that point:

$$\begin{aligned}
\phi(s) &= \tfrac{1}{2}((1-s)^2x_1^2 + b(1-bs)^2x_2^2), && \text{(definition of } f\text{)}\\
\phi'(s) &= -(1-s)x_1^2 - b^2(1-bs)x_2^2 && \text{(chain rule, term by term)}\\
&= -x_1^2 + s\,x_1^2 - b^2x_2^2 + b^3s\,x_2^2 && \text{(expand the products)}\\
&= s\,(x_1^2 + b^3x_2^2) - (x_1^2 + b^2x_2^2). && \text{(collect the terms in } s\text{)}
\end{aligned}$$

For $\underset{\sim}{x} \neq \underset{\sim}{0}_{2\times 1}$, $x_1^2 + b^3x_2^2 > 0$, so ϕ' is increasing in s with a single zero, and ϕ is minimized at $s^* = \dfrac{x_1^2 + b^2x_2^2}{x_1^2 + b^3x_2^2}$. From $\underset{\sim}{x}^{(0)} = (b, 1)^\top$:

$$\begin{aligned}
s^{(0)} &= \frac{b^2 + b^2}{b^2 + b^3} && \text{(substitute } x_1 = b, x_2 = 1\text{)}\\
&= \frac{2}{1+b}, && \text{(cancel } b^2\text{)}
\end{aligned}$$

and, with $r \overset{\text{def}}{=} \frac{b-1}{b+1}$,

$$\begin{aligned}
1 - s^{(0)} &= \frac{1+b-2}{1+b}\\
&= r, && \text{(common denominator)}\\
1 - b\,s^{(0)} &= \frac{1+b-2b}{1+b}\\
&= -r, && \text{(common denominator)}\\
\underset{\sim}{x}^{(1)} &= (r\,b,\ -r\cdot 1)^\top\\
&= r\,(b, -1)^\top. && \text{(the gradient step from } (b,1)^\top\text{)}
\end{aligned}$$

For $b \neq 1$ the same holds at every step, by induction (when $b = 1$, $r = 0$ and $\underset{\sim}{x}^{(1)}$ is already the minimizer): if $\underset{\sim}{x}^{(t)} = (c\,b,\ \pm c)^\top$ for a number $c \neq 0$, then s^*, which depends only on $x_1^2 = c^2b^2$ and $x_2^2 = c^2$, is again $\frac{2}{1+b}$ (the factor c^2 cancels), and the step multiplies the first entry by r and the second by $-r$. So

$$\begin{aligned} \underset{\sim}{x}^{(t)} &= (b\,r^t,\ (-r)^t)^\top, \\ f(\underset{\sim}{x}^{(t)}) &= \tfrac{1}{2}(b^2+b)\,r^{2t} \\ &= r^{2t} f(\underset{\sim}{x}^{(0)}). \end{aligned}$$

For $b \neq 1$, exactly one entry changes sign at every step: the second when $b > 1$ (then $r > 0$), and the first when $b < 1$ (then $r < 0$). So the iterates zigzag toward the minimizer $\underset{\sim}{0}_{2\times 1}$.

- With $b = 3$, $r = \frac{1}{2}$: $\underset{\sim}{x}^{(1)} = (1.5, -0.5)^\top$ and $\underset{\sim}{x}^{(2)} = (0.75, 0.25)^\top$, and f falls from 6 to 1.5 to 0.375, a quarter at each step.
- With $b = 1$, $r = 0$ and one step reaches the minimizer.
- With $b = 0.01$, $r^2 = (0.99/1.01)^2 \approx 0.96$, so f shrinks by only about 4% per step. Newton's method reaches the minimizer of any such f in one step (Theorem 2 in Newton's Method).

Definition 0.6 (Rosenbrock function)**.** The **Rosenbrock function** on $\mathbb{R}^2$ is

$$f(\underset{\sim}{x}) \stackrel{\text{def}}{=} 100(x_2 - x_1^2)^2 + (1 - x_1)^2, \qquad \underset{\sim}{x} = (x_1, x_2)^\top.$$

It is a standard test problem for optimizers [3].

Example 0.7 (Values of the Rosenbrock function)**.** We evaluate the Rosenbrock function at the minimizer $(1, 1)$, at the origin, and at the standard starting point $(-1.2, 1)$. Then we evaluate it along the curve $x_2 = x_1^2$, which is the floor of a narrow, curved valley.

R

```r
rosenbrock_rf <- function(x1, x2) 100 * (x2 - x1^2)^2 + (1 - x1)^2

points_rf <- list(minimizer = c(1, 1), origin = c(0, 0), start = c(-1.2, 1))
sapply(points_rf, function(p) round(rosenbrock_rf(p[1], p[2]), 3))
#> minimizer    origin     start
#>       0.0       1.0      24.2

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf <- c(-1, 0, 0.5, 1, 2)
rbind(
  on_curve = rosenbrock_rf(x_curve_rf, x_curve_rf^2),
  formula = (1 - x_curve_rf)^2
)
#>          [,1] [,2] [,3] [,4] [,5]
#> on_curve    4    1 0.25    0    1
#> formula     4    1 0.25    0    1
```

Python

```python
import numpy as np

def rosenbrock_rf(x1, x2):
    return 100 * (x2 - x1 ** 2) ** 2 + (1 - x1) ** 2

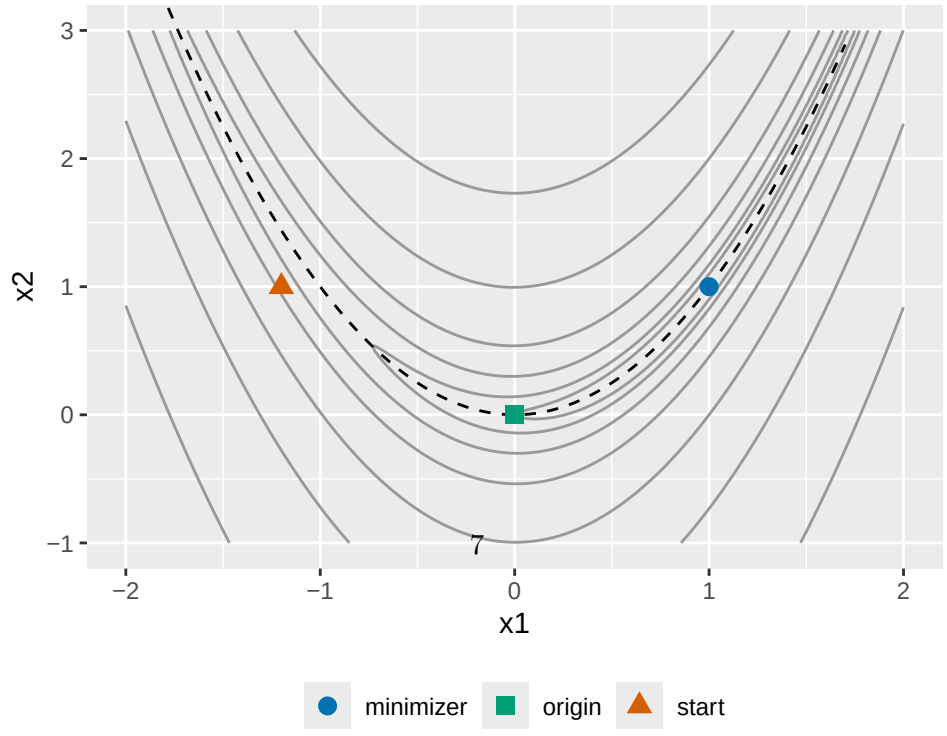
points_rf = {"minimizer": (1, 1), "origin": (0, 0), "start": (-1.2, 1)}
print({name: round(rosenbrock_rf(*p), 3) for name, p in points_rf.items()})
#> {'minimizer': 0, 'origin': 1, 'start': 24.2}

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf = np.array([-1.0, 0.0, 0.5, 1.0, 2.0])
print("on_curve:", rosenbrock_rf(x_curve_rf, x_curve_rf ** 2))
#> on_curve: [4.   1.   0.25 0.   1.  ]
print("formula: ", (1 - x_curve_rf) ** 2)
#> formula:  [4.   1.   0.25 0.   1.  ]
```

The function is 0 at $(1, 1)$ and positive everywhere else, because both of its parts are non-negative and vanish together only when $x_1 = 1$ and $x_2 = 1$. So $(1, 1)$ is the global minimizer. At the starting point the value is 24.2, mostly from the first term, because $(-1.2, 1)$ is off the curve $x_2 = x_1^2$. On the curve the function is $(1 - x_1)^2$, which varies slowly compared with the steep walls.

R

```
library(ggplot2)
rf_grid <- expand.grid(
  x1 = seq(-2, 2, length.out = 201),
  x2 = seq(-1, 3, length.out = 201)
)
rf_grid$f <- rosenbrock_rf(rf_grid$x1, rf_grid$x2)
rf_points <- data.frame(
  name = names(points_rf),
  x1 = sapply(points_rf, `[`, 1),
  x2 = sapply(points_rf, `[`, 2)
)
rf_curve <- data.frame(x1 = seq(-2, 1.7, length.out = 200))
rf_curve$x2 <- rf_curve$x1^2
ggplot(rf_grid, aes(x1, x2)) +
  geom_contour(
    aes(z = f),
    colour = "grey60",
    breaks = c(1, 3, 10, 30, 100, 300, 1000)
  ) +
  geom_line(data = rf_curve, linetype = "dashed") +
  geom_point(aes(colour = name, shape = name), data = rf_points, size = 3) +
  scale_colour_manual(
    values = c(minimizer = "#0072B2", origin = "#009E73", start = "#D55E00")
  ) +
  scale_shape_manual(values = c(minimizer = 16, origin = 15, start = 17)) +
  coord_cartesian(xlim = c(-2, 2), ylim = c(-1, 3)) +
  labs(x = "x1", y = "x2", colour = NULL, shape = NULL) +
  theme(legend.position = "bottom")
```



Definition 0.7 (Basin of attraction)**.** For given step lengths, the **basin of attraction** of a local minimizer $\underset{\sim}{x}^*$ under gradient descent is the set of starting points $\underset{\sim}{x}^{(0)}$ from which the iterates converge to $\underset{\sim}{x}^*$ [4].

Example 0.8 (Basins of a double-well function)**.** Let $f(x) = x^4 - 2x^2$, so $\frac{\partial}{\partial x} f(x) = 4x(x-1)(x+1)$ and the stationary points are -1, 0 and 1. Both -1 and 1 are local minimizers, with $f(\pm 1) = -1$, and 0 is a local maximizer. We run gradient descent with the fixed step length $s = 0.05$ for 200 steps from the starting points -1.5, -0.5, -0.1, 0.1, 0.5, 1.5 and 3, and for three steps from $x^{(0)} = 4$.

R

```r
basin_step_bs <- 0.05
basin_starts_bs <- c(-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3)

# 200 steps from every starting point at once.
basin_final_bs <- basin_starts_bs
for (i in 1:200) {
  basin_slope_bs <- 4 * basin_final_bs^3 - 4 * basin_final_bs
  basin_final_bs <- basin_final_bs - basin_step_bs * basin_slope_bs
}
basin_end_bs <- signif(basin_final_bs, 3)
tibble::tibble(start = basin_starts_bs, end = basin_end_bs)
#> # A tibble: 7 x 2
#>   start   end
#>   <dbl> <dbl>
#> 1  -1.5    -1
#> 2  -0.5    -1
#> 3  -0.1    -1
#> 4   0.1     1
#> 5   0.5     1
#> 6   1.5     1
#> 7   3      -1

# The first step from 3.
basin_from_three_bs <- 3 - basin_step_bs * (4 * 3^3 - 4 * 3)

# Three steps from 4: the iterates grow in size.
basin_far_bs <- numeric(4)
basin_far_bs[1] <- 4
for (i in 1:3) {
  basin_slope_bs <- 4 * basin_far_bs[i]^3 - 4 * basin_far_bs[i]
  basin_far_bs[i + 1] <- basin_far_bs[i] - basin_step_bs * basin_slope_bs
}
signif(basin_far_bs, 3)
#> [1]       4.0      -8.0      92.8 -160000.0
```

Python

```
basin_step_bs = 0.05

def basin_next_bs(x):
    return x - basin_step_bs * (4 * x ** 3 - 4 * x)

def basin_gd_bs(x0, n_steps=200):
    x = x0
    for _ in range(n_steps):
        x = basin_next_bs(x)
    return x

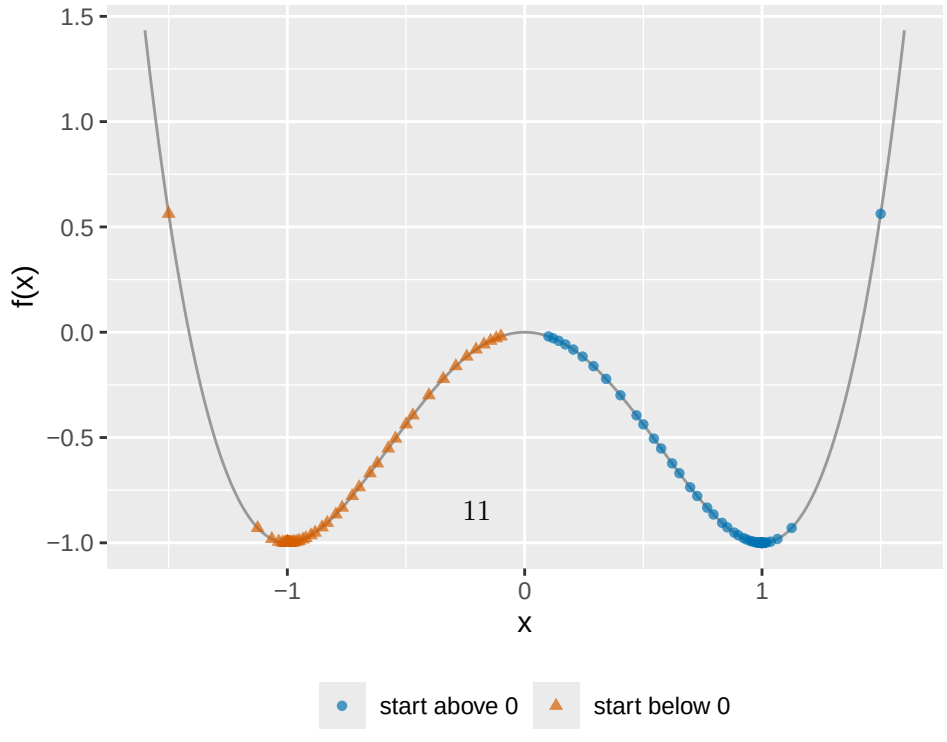
basin_starts_bs = [-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3]
for start in basin_starts_bs:
    print(start, float(f"{basin_gd_bs(start):.3g}"))
#> -1.5 -1.0
#> -0.5 -1.0
#> -0.1 -1.0
#> 0.1 1.0
#> 0.5 1.0
#> 1.5 1.0
#> 3 -1.0

# Three steps from 4: the iterates grow in size.
basin_far_bs = [4.0]
for _ in range(3):
    basin_far_bs.append(basin_next_bs(basin_far_bs[-1]))
print([float(f"{v:.3g}") for v in basin_far_bs])
#> [4.0, -8.0, 92.8, -160000.0]
```

Each of the six starts between -1.5 and 1.5 ends at -1 or 1, whichever lies on its own side of 0, so for these starts the sign of the starting point decides which of the two the run reaches. A run from 0 itself never moves, because $\frac{\partial}{\partial x} f(0) = 0$. The basin of attraction of 1 is not all of $(0, \infty)$: the first step from $x^{(0)} = 3$ lands at -1.8, on the other side of 0, and the run ends at -1. From $x^{(0)} = 4$ the iterates, starting with $x^{(0)}$ itself, are 4, -8, 92.8 and -1.6×10^5, which grow in absolute value at every step shown. Both of these starts are far from 0, where the derivative is large and so the step $s \frac{\partial}{\partial x} f(x)$ is long. From $x^{(0)} = 3$ that step is $96\,s$ long, so the first step stays on the positive side of 0 only when s is below 0.03125. So the basin of 1 depends on s, and with $s = 0.05$ the basin is not a half-line.

R

```
# Uses basin_step_bs and basin_starts_bs from the example above.
basin_runs_bf <- matrix(0, nrow = 31, ncol = 6)
basin_runs_bf[1, ] <- basin_starts_bs[1:6]
for (i in 1:30) {
  basin_prev_bf <- basin_runs_bf[i, ]
  basin_slope_bf <- 4 * basin_prev_bf^3 - 4 * basin_prev_bf
  basin_runs_bf[i + 1, ] <- basin_prev_bf - basin_step_bs * basin_slope_bf
}
basin_side_bf <- ifelse(
  basin_starts_bs[1:6] > 0, "start above 0", "start below 0"
)
basin_x_bf <- as.vector(basin_runs_bf)
basin_side_all_bf <- rep(basin_side_bf, each = 31)
basin_paths_bf <- tibble::tibble(x = basin_x_bf, side = basin_side_all_bf)
basin_paths_bf$cost <- basin_paths_bf$x^4 - 2 * basin_paths_bf$x^2
basin_grid_bf <- seq(-1.6, 1.6, length.out = 321)
basin_curve_bf <- tibble::tibble(
  x = basin_grid_bf, cost = basin_grid_bf^4 - 2 * basin_grid_bf^2
)
basin_colours_bf <- c("start above 0" = "#0072B2", "start below 0" = "#D55E00")
ggplot2::ggplot(basin_curve_bf, ggplot2::aes(x, cost)) +
  ggplot2::geom_line(colour = "grey60") +
  ggplot2::geom_point(
    data = basin_paths_bf,
    ggplot2::aes(colour = side, shape = side), alpha = 0.7
  ) +
  ggplot2::scale_colour_manual(values = basin_colours_bf) +
  ggplot2::labs(x = "x", y = "f(x)", colour = NULL, shape = NULL) +
  ggplot2::theme(legend.position = "bottom")
```



Proposition 0.1 (Divergence of gradient descent on the double-well function)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. If* $x^{(0)}$ *satisfies*

$$x^{(0)^2} > 1 + \frac{1}{2s},$$

then $\left|x^{(t)}\right| \to \infty$ *as* $t \to \infty$*.*

Proof

Proof. The derivative is $\frac{\partial}{\partial x} f(x) = 4x^3 - 4x$, so one step of gradient descent gives

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - s\left(4x^{(t)^3} - 4x^{(t)}\right) && \text{(gradient descent step)} \\ &= -x^{(t)}\left(4sx^{(t)^2} - 4s - 1\right). && \text{(factor out } -x^{(t)}\text{)} \end{aligned}$$

Write $m(x) \stackrel{\text{def}}{=} 4sx^2 - 4s - 1$. Then $\left|x^{(t+1)}\right| = \left|x^{(t)}\right| m(x^{(t)})$ whenever $m(x^{(t)}) \geq 0$. Now $m(x) > 1$ exactly when $x^2 > 1 + 1/(2s)$, and $m(x)$ increases with x^2. So if $x^{(t)}$ satisfies the condition, then $\left|x^{(t+1)}\right| > \left|x^{(t)}\right|$, and $x^{(t+1)}$ satisfies it too. By induction on t, every iterate satisfies it. Since $\left|x^{(t)}\right|$ increases at every step, $x^{(t)^2} \geq x^{(0)^2}$, and so $m(x^{(t)}) \geq m(x^{(0)}) > 1$. Hence $\left|x^{(t)}\right| \geq m(x^{(0)})^t \left|x^{(0)}\right|$, which tends to ∞. □

Corollary 0.1 (Basins of the double-well function are bounded)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. Every start* $x^{(0)}$ *in the basin of attraction of a local minimizer of* f *satisfies*

$$x^{(0)^2} \leq 1 + \frac{1}{2s}.$$

Proof

Proof. The iterates from a start in the basin converge to a local minimizer, so they do not diverge to ∞. By Proposition 0.1, the start therefore does not satisfy $x^{(0)^2} > 1 + 1/(2s)$. □

Example 0.9 (Divergent starts on the double-well function for three step lengths)**.** For $f(x) = x^4 - 2x^2$, Proposition 0.1 says that gradient descent diverges to ∞ from every start with $\left|x^{(0)}\right|$ above $\sqrt{1 + 1/(2s)}$. This bound is 7.14 for $s = 0.01$, 3.32 for $s = 0.05$, and 2.45 for $s = 0.1$. The derivative $4x^3 - 4x$ is unbounded, as is the second derivative $12x^2 - 4$, so the step $s\frac{\partial}{\partial x} f(x)$ grows without bound with x, and a far start overshoots. By Proposition 0.1, every fixed step length fails to converge from some starts. A larger step length lowers the bound, so more starts are guaranteed to diverge. By Corollary 0.1,

each basin of attraction lies within the same bounded interval, which narrows as s grows. In the earlier basin example, $s = 0.05$, and the start 4 lies above the bound 3.32, which is why its iterates grow.

Definition 0.8 (Plateau)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ be a function with a gradient at every point of a region, and let $\epsilon > 0$ be a tolerance. The region is a **plateau** of f if the Euclidean norm of the gradient satisfies

$$0 < \left\| \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \right\| < \epsilon$$

at every point $\underset{\sim}{x}$ of the region. The function is then nearly flat on the region, and the region contains no stationary point.

Example 0.10 (A cost with no stationary point)**.**
Let $f(x) = \log(1 + \mathrm{e}^{-x})$. Its derivative is $\frac{\partial}{\partial x} f(x) = -1/(1 + \mathrm{e}^{x})$, which is negative for every x, so f has no stationary point at all.
At $x = 0$ the gradient has size 0.5. At $x = 10$ it has size 4.54×10^{-5}, which is small but not zero. A step of gradient descent subtracts the gradient times the step length. The gradient is negative, so the iterates keep moving to the right, more and more slowly. With tolerance $\epsilon = 10^{-4}$, the region $x \geq 10$ is a plateau (Definition 0.8). A run that enters this plateau has not converged; the run has only slowed down.

Definition 0.9 (Armijo condition)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x} \in \mathbb{R}^p$, and let $g \overset{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$. A step length $s > 0$ satisfies the **Armijo condition** [5] with constant $c \in (0, 1)$ at $\underset{\sim}{x}$ if

$$f(\underset{\sim}{x} - s\underset{\sim}{g}) \leq f(\underset{\sim}{x}) - c\, s\, \|\underset{\sim}{g}\|^2.$$

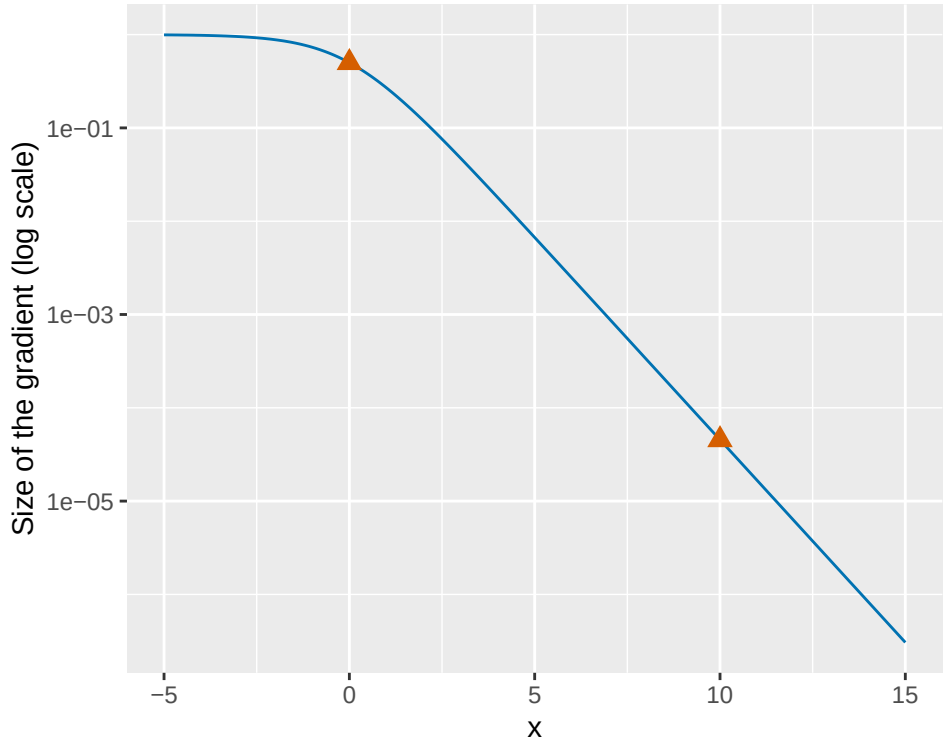
Example 0.11 (The Armijo condition on a parabola)**.** Let $f(x) = x^2$ and $x = 2$, so $g = f'(2) = 4$ and $\|g\|^2 = 16$. With $c = 1/4$, the condition on a step length $s > 0$ is

$$\begin{aligned} (2 - 4s)^2 &\leq 4 - 4s && (f(2 - 4s) \leq f(2) - c\, s\, \|g\|^2) \\ 4 - 16s + 16s^2 &\leq 4 - 4s && (\text{expand the square}) \\ 16s^2 &\leq 12s && (\text{add } 16s - 4 \text{ to both sides}) \\ s &\leq 3/4. && (\text{divide by } 16s > 0) \end{aligned}$$

So $s = 1/2$ satisfies the condition: $f(0) = 0 \leq 4 - 2 = 2$. And $s = 1$ does not: $f(-2) = 4 > 4 - 4 = 0$.

R

```
plateau_grid_pf <- seq(-5, 15, length.out = 401)
plateau_exp_pf <- exp(plateau_grid_pf)
plateau_size_pf <- 1 / (1 + plateau_exp_pf)
plateau_curve_pf <- tibble::tibble(x = plateau_grid_pf, size = plateau_size_pf)
plateau_marks_x_pf <- c(0, 10)
plateau_marks_exp_pf <- exp(plateau_marks_x_pf)
plateau_marks_size_pf <- 1 / (1 + plateau_marks_exp_pf)
plateau_marks_pf <- tibble::tibble(
  x = plateau_marks_x_pf, size = plateau_marks_size_pf
)
plateau_aes_pf <- ggplot2::aes(x, size)
ggplot2::ggplot(plateau_curve_pf, plateau_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(
    data = plateau_marks_pf, colour = "#D55E00", shape = 17, size = 3
  ) +
  ggplot2::scale_y_log10() +
  ggplot2::labs(x = "x", y = "Size of the gradient (log scale)")
```



Python

```
import matplotlib.pyplot as plt
import numpy as np

plateau_grid_pf = np.linspace(-5, 15, 401)
plateau_marks_pf = np.array([0, 10])
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(plateau_grid_pf, 1 / (1 + np.exp(plateau_grid_pf)), color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7f7864353250>]
```

Definition 0.10 (Backtracking line search)**.** Let $s_0 > 0$ be an initial step length, $\beta \in (0,1)$ a shrink factor, and $c \in (0,1)$. **Backtracking line search** [5] is the line search that chooses $s^{(t)}$ to be the largest of $s_0, \beta s_0, \beta^2 s_0, \ldots$ that satisfies the Armijo condition with constant c at $\underset{\sim}{x}^{(t)}$.

Example 0.12 (Backtracking on a parabola)**.** For $f(x) = x^2$ at $x = 2$ with $c = 1/4$, the Armijo condition holds exactly when $s \le 3/4$ (Example 0.11). With $s_0 = 1$ and $\beta = 1/2$, the trial step lengths are $1, 1/2, 1/4, \ldots$. The first, $s = 1$, exceeds $3/4$ and is rejected. The second, $s = 1/2$, is at most $3/4$ and is accepted. So backtracking chooses $s^{(0)} = 1/2$, after two evaluations of f.

Theorem 0.2 (A bound on the error of gradient descent on a quadratic)**.** *Let* $\mathbf{S}$ *be a* $p \times p$ *positive definite matrix with eigenvalues* $\lambda_1, \ldots, \lambda_p$*, let* $\underset{\sim}{c} \in \mathbb{R}^p$*, and let* $f(\underset{\sim}{x}) = \frac{1}{2}\,\underset{\sim}{x}^\top \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c}\cdot\underset{\sim}{x}$*, whose unique minimizer is* $\underset{\sim}{x}^* = \mathbf{S}^{-1}\underset{\sim}{c}$ *(Theorem 2 in Newton's Method). Gradient descent with a fixed step length* $s > 0$ *from any* $\underset{\sim}{x}^{(0)}$ *has errors satisfying, for* $t = 1, 2, \ldots,$

$$\left\|\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*\right\| \le \rho(s)^t \left\|\underset{\sim}{x}^{(0)} - \underset{\sim}{x}^*\right\|, \qquad \rho(s) \overset{def}{=} \max_{1 \le i \le p} |1 - s\,\lambda_i|.$$

Proof

Proof. Let $\underset{\sim}{e}^{(t)} \overset{\text{def}}{=} \underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*$ be the error at step t. The gradient of f is $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c}$ (see the proof of Theorem 2 in Newton's Method), and $\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*$, so

$$\begin{aligned}
\underset{\sim}{e}^{(t+1)} &= \underset{\sim}{x}^{(t)} - s\,(\mathbf{S}\underset{\sim}{x}^{(t)} - \underset{\sim}{c}) - \underset{\sim}{x}^* && \text{(definition of gradient descent)} \\
&= \underset{\sim}{x}^{(t)} - s\,\mathbf{S}(\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*) - \underset{\sim}{x}^* && (\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*) \\
&= (\mathbf{I}_p - s\,\mathbf{S})\,\underset{\sim}{e}^{(t)}. && \text{(collect the terms in } \underset{\sim}{e}^{(t)})
\end{aligned}$$

By induction on t, $\underset{\sim}{e}^{(t)} = (\mathbf{I}_p - s\,\mathbf{S})^t\,\underset{\sim}{e}^{(0)}$. Write $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ (spectral theorem). Then $\mathbf{I}_p - s\,\mathbf{S} = \mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})\mathbf{Q}^\top$ because $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$, and the tth power cancels each inner $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_p$, so

$$\begin{aligned}
\left\|\underset{\sim}{e}^{(t)}\right\| &= \left\|\mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})^t\mathbf{Q}^\top\underset{\sim}{e}^{(0)}\right\| && \text{(the factorization above)} \\
&= \left\|(\mathbf{I}_p - s\,\mathbf{\Lambda})^t\mathbf{Q}^\top\underset{\sim}{e}^{(0)}\right\| && \text{(orthogonal matrices preserve length)} \\
&\le \rho(s)^t \left\|\mathbf{Q}^\top\underset{\sim}{e}^{(0)}\right\| && \text{(each coordinate is multiplied by } (1 - s\lambda_i)^t) \\
&= \rho(s)^t \left\|\underset{\sim}{e}^{(0)}\right\|. && \text{(orthogonal matrices preserve length)}
\end{aligned}$$

The length-preserving steps apply orthogonal matrices preserve length. The inequality holds because $\mathbf{I}_p - s\,\mathbf{\Lambda}$ is diagonal, so its tth power multiplies coordinate i by $(1 - s\,\lambda_i)^t$, whose absolute value is at most $\rho(s)^t$. □

Example 0.13 (The error multiplier for a step length of one quarter)**.** Let $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1 (eigenvalue example), and $\underset{\sim}{c} = \underset{\sim}{0}_{2\times 1}$, so $\underset{\sim}{x}^* = \underset{\sim}{0}_{2\times 1}$. With $s = 0.25$, the factors $1 - s\,\lambda_i$ are 0.25 and 0.75, so $\rho(s) = 0.75$ and

$$\left\|\underset{\sim}{x}^{(t)}\right\| \le 0.75^t \left\|\underset{\sim}{x}^{(0)}\right\|.$$

Along the eigenvector for $\lambda = 3$ the error shrinks fast, and along the eigenvector for $\lambda = 1$ it shrinks slowly; the bound follows the slower one.

Theorem 0.3 (The best constant step length)**.** *In the setting of Theorem 0.2, let* $\lambda_{\min}$ *and* $\lambda_{\max}$ *be the smallest and largest eigenvalues of* $\mathbf{S}$*. The factor* $\rho(s) = \max_i |1 - s\,\lambda_i|$ *of Theorem 0.2 is smallest over* $s > 0$ *at*

$$s^* = \frac{2}{\lambda_{\min} + \lambda_{\max}},$$

where

$$\rho(s^*) = \frac{\kappa - 1}{\kappa + 1}, \qquad \kappa = \kappa(\mathbf{S}),$$

the condition number of $\mathbf{S}$*.*

Proof

Proof. For $s > 0$, each $1 - s\,\lambda_i$ lies between $u_{\min} \overset{\text{def}}{=} 1 - s\,\lambda_{\max}$ and $u_{\max} \overset{\text{def}}{=} 1 - s\,\lambda_{\min}$, so $|1 - s\,\lambda_i| \le \max(u_{\max}, -u_{\min})$. Both $\lambda_{\min}$ and $\lambda_{\max}$ occur among the λ_i, and $|u_{\max}| \ge u_{\max}$ and $|u_{\min}| \ge -u_{\min}$, so the maximum is attained:

$$\rho(s) = \max(1 - s\,\lambda_{\min},\ s\,\lambda_{\max} - 1).$$

The first expression is strictly decreasing in s, and the second is strictly increasing, and they coincide when

$$\begin{aligned} 1 - s\,\lambda_{\min} &= s\,\lambda_{\max} - 1 && \text{(set the two expressions equal)} \\ 2 &= s\,(\lambda_{\min} + \lambda_{\max}) && \text{(collect the terms in } s\text{)} \\ s &= \frac{2}{\lambda_{\min} + \lambda_{\max}} = s^*. && \text{(divide)} \end{aligned}$$

For $s < s^*$, $\rho(s)$ is at least the decreasing expression, which exceeds its value at s^*. For $s > s^*$, $\rho(s)$ is at least the increasing expression, which exceeds its value at s^*. So s^* is the minimizer, and

$$\begin{aligned}
\rho(s^*) &= 1 - \frac{2\,\lambda_{\min}}{\lambda_{\min} + \lambda_{\max}} && \text{(first expression at } s^*\text{)} \\
&= \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} && \text{(common denominator)} \\
&= \frac{\kappa - 1}{\kappa + 1}. && \text{(divide the numerator and denominator by } \lambda_{\min}\text{)}
\end{aligned}$$

□

Example 0.14 (The best step length for eigenvalues 3 and 1)**.** For $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1, Theorem 0.3 gives $s^* = 2/(3+1) = 0.5$, $\kappa = 3$ (condition number example), and $\rho(s^*) = (3-1)/(3+1) = 0.5$. A grid search over $s = 0.01, 0.02, \ldots, 1$ finds the smallest $\rho(s)$ at $s = 0.5$, the same step length. Each step then multiplies the error by at most 0.5.

Exercise 0.1 (Which step lengths shrink the error bound)**.** In the setting of Theorem 0.2, find all $s > 0$ with $\rho(s) < 1$, and write the answer as a condition on s that involves $\lambda_{\max}$.

Solution

Solution 0.1. $\rho(s) < 1$ says that $|1 - s\,\lambda_i| < 1$ for every i, which is $-1 < 1 - s\,\lambda_i < 1$. The right inequality holds for every $s > 0$ because $\lambda_i > 0$. The left inequality is $s\,\lambda_i < 2$, that is, $s < 2/\lambda_i$. These p inequalities, one for each eigenvalue, all hold exactly when the inequality for the largest eigenvalue holds, since it is the most restrictive:

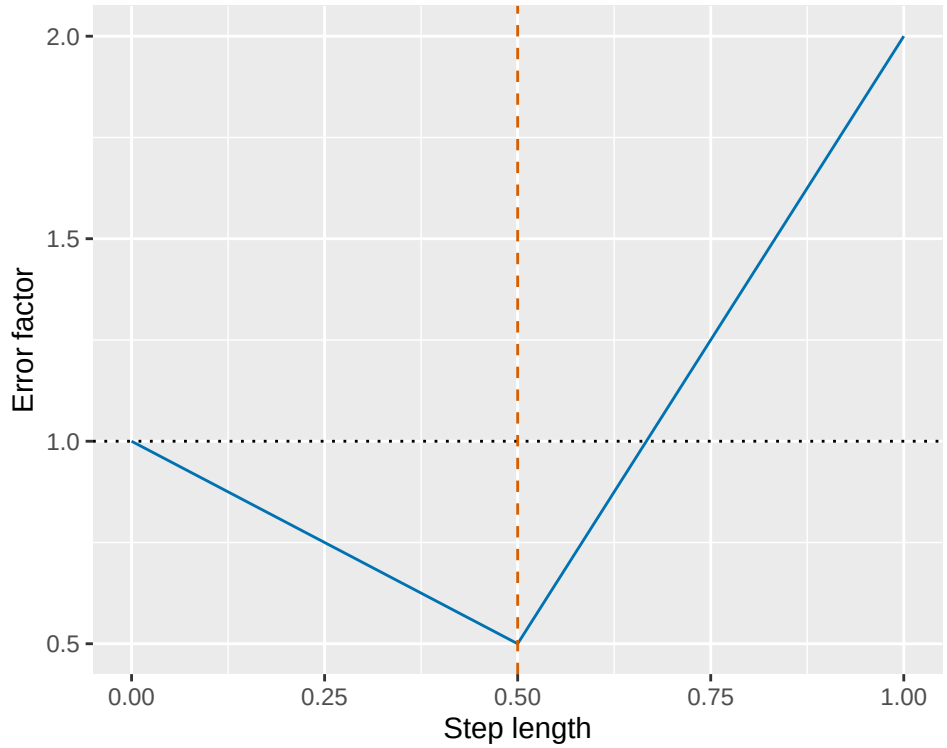
$$0 < s < \frac{2}{\lambda_{\max}}.$$

Exercise 0.2 (The best step length for eigenvalues 1 and 9)**.** Gradient descent with a fixed step length converges slowly if the step is too short and diverges if it is too long. Theorem 0.3 gives the step length that shrinks the error bound fastest, from the smallest and largest eigenvalues of $\mathbf{S}$.

Let $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$, a diagonal matrix, so its eigenvalues are its diagonal entries, 1 and 9. Using Theorem 0.3, find s^*.

R

```
bcs_grid_pf <- seq(0, 1, length.out = 401)
bcs_first_pf <- abs(1 - 3 * bcs_grid_pf)
bcs_second_pf <- abs(1 - bcs_grid_pf)
bcs_rho_pf <- pmax(bcs_first_pf, bcs_second_pf)
bcs_curve_pf <- tibble::tibble(s = bcs_grid_pf, rho = bcs_rho_pf)
bcs_aes_pf <- ggplot2::aes(s, rho)
ggplot2::ggplot(bcs_curve_pf, bcs_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_hline(yintercept = 1, linetype = "dotted") +
  ggplot2::geom_vline(
    xintercept = 0.5, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::labs(x = "Step length", y = "Error factor")
```



Python

```
import matplotlib.pyplot as plt
import numpy as np

bcs_grid_pf = np.linspace(0, 1, 401)
bcs_rho_pf = np.maximum(np.abs(1 - 3 * bcs_grid_pf), np.abs(1 - bcs_grid_pf))
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(bcs_grid_pf, bcs_rho_pf, color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7f78620596d0>]
ax.axhline(1, color="black", linestyle=":")
#> <matplotlib.lines.Line2D object at 0x7f7862059810>
ax.axvline(0.5, color="#D55E00", linestyle="--")
#> <matplotlib.lines.Line2D object at 0x7f7862059950>
ax.set_xlabel("Step length")
```

Solution

Solution 0.2. $\mathbf{S}$ is diagonal, so its eigenvalues are its diagonal entries 1 and 9. Then

$$s^* = \frac{2}{1+9} = 0.2.$$

Exercise 0.3 (The best error factor for eigenvalues 1 and 9)**.** In the setting of Exercise 0.2, find $\rho(s^*)$.

Solution

Solution 0.3. The condition number is $\kappa(\mathbf{S}) = 9/1 = 9$, so by Theorem 0.3

$$\rho(s^*) = \frac{9-1}{9+1} = 0.8.$$

Exercise 0.4 (Steps to cut the error bound by a factor of ten)**.** In the setting of Exercise 0.2 and Exercise 0.3, find the smallest $t = 1, 2, \ldots$ with $\rho(s^*)^t \leq 0.1$.

Solution

Solution 0.4. From Solution 0.3, $\rho(s^*) = 0.8$. Taking logarithms, $\rho(s^*)^t \leq 0.1$ becomes $t \ln \rho(s^*) \leq \ln 0.1$. Here $\ln \rho(s^*) < 0$, so dividing by it flips the inequality: $t \geq \ln 0.1 / \ln \rho(s^*) \approx 10.32$. The smallest such t is 11.

[1] Wikipedia contributors, "Gradient descent — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Gradient_descent

[2] Wikipedia contributors, "Line search — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Line_search

[3] Wikipedia contributors, "Rosenbrock function — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Rosenbrock_function

[4] Wikipedia contributors, "Attractor — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Attractor

[5] Wikipedia contributors, "Backtracking line search — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Backtracking_line_search